

United International University

Name: MD ABDULLAH AL SADIK ID: 0112430281Course Code: _____ Section X Group: 4 Trimester: _____ Date: _____

Experiment No. 04

Name of the Experiment: Determination of the spring constant and effective mass of a given spiral spring.

Theory:

If a spring is clamped vertically at the end P, and loaded with a mass m_0 at the other end A, then the period of vibration of the spring along a vertical line is given by,

$$T = 2\pi \sqrt{\frac{m' + m_0}{k}} = 2\pi \sqrt{\frac{M}{k}} \dots \dots \dots (1)$$

Where, m' is a constant called the effective mass of the spring and k , the spring constant i.e. the ratio between the added force and the corresponding stretch of the spring.

The contribution of the mass of the spring to the effective mass of the vibrating system can be shown as follows: Consider the kinetic energy of a spring and its load undergoing simple harmonic motion. At the instant under consideration, let the load m_0 be moving with velocity v_0 as shown in the figure.

At the same instant, an element dm of the mass m of the spring will also be moving up but with a velocity v which is smaller than v_0 . It is evident that the ratio between v and v_0 is just the ratio between y and y_0 . Hence, $\frac{v}{y} = \frac{v_0}{y_0}$ i.e.

$v = \frac{v_0}{y_0} y$. The kinetic energy of the spring alone

will be $\int \frac{v^2}{2} dm$. But dm may be written as $\frac{m}{y_0} dy$,

where, m is the mass of the spring.

Thus the integral equals to $\frac{1}{2} \left(\frac{m}{3} \right) v_0^2$. The total

kinetic energy of the system will then be $\frac{1}{2} \left(m_0 + \frac{m}{3} \right) v_0^2$ and the total mass of the system is

therefore, $M = \left(m_0 + \frac{m}{3} \right) = \left(M_0 + m_0 + \frac{m}{3} \right)$ where M_0 is the initial load or mass of the stand, if used.

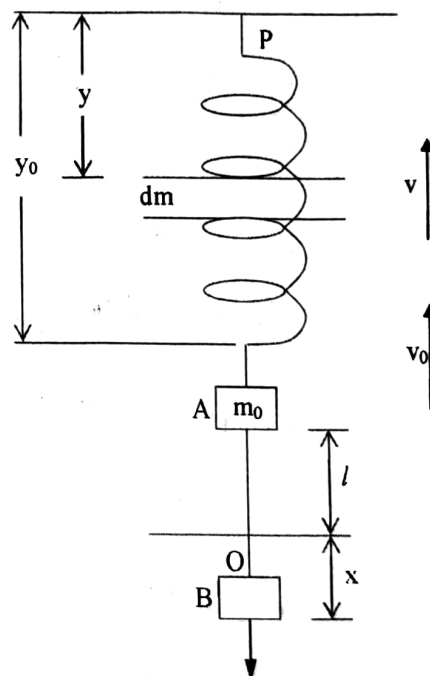


Fig. 01

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Hence, effective mass, $m' = \frac{m}{3}$. The applied force m_0g [or $(m_0 + M_0)g$ if used] is proportional to the extension l within the elastic limit.

$$\text{Therefore, } mg = kl \text{ or, } l = \frac{g}{k} m$$

$$k = \frac{g}{l/m} = \frac{g}{\text{slope of } l \text{ vs } m \text{ graph}} =$$

Apparatus:

- A spiral spring
- Convenient masses with hanging arrangement
- Clamp or a hook attached to a rigid framework of heavy metal rods
- Weighing balance
- Stopwatch and scale

Experimental Data:

(A) Initial Length of the Spring, $L_0 = 12 \text{ cm}$

(B) Initial Load (mass of the stand), $M_0 = 150 \text{ gm}$

(C) Mass of the spring, $m = 30 \text{ gm}$

(D) Table for determining extensions and time periods:

No. of Obs.	Added Loads, m_0 (gm.)	Length of the Spring, L (cm)	Extension, $l = L - L_0$ (cm.)	No. of vibrations	Total time (sec.)	Mean time (sec.)	Period, T (sec.)	T^2 (sec. ²)
1	50	15 cm	3 cm	10	6.25	6.175	0.6175	0.3813
					6.10			
2	100	18 cm	6 cm	10	7.29	7.24	0.724	0.5242
					7.19			
3	150	22 cm	10 cm	10	8.38	8.4	0.84	0.7056
					8.42			
4	200	25 cm	13 cm	10	9.20	9.15	0.915	0.8372
					9.10			
5	250	28 cm	16 cm	10	9.80	9.855	0.9855	0.9712
					9.91			

Calculation:

(A) Effective mass, m' (Experimental effective mass from graph T^2 vs. m_0) = 50 gm

(B) Effective mass, m' (Theoretical effective mass from calculation) = 43.33 gm

(C) Difference (%) = $\frac{\text{Experimental Result} - \text{Theoretical Result}}{\text{Theoretical Result}} \times 100\% = 15.34\%$

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(D) Accuracy (%) = $100\% - \% \text{ Difference} =$

(E) Spring constant, k from l vs m_0 graph:

$$\therefore \text{Spring Constant } k = \frac{g}{\text{slope of } l \text{ vs } m_0 \text{ graph}}$$

$$= \frac{980}{0.675}$$

$$= 145185 \text{ dynes/cm}$$

Slope = slope of l vs $m_0 = l/m_0$

$$= \frac{l_2 - l_1}{m_{02} - m_{01}} = \frac{13.2 - 7.8}{200 - 120}$$

$$= 0.675$$

Results:

(A) Value of the Effective Mass, $m' = 50 \text{ gm}$

(B) Value of the Spring Constant, $k = 145185 \text{ dynes/cm}$

Discussions:

Q: What do you understand by the term *Spring Constant*?

The spring constant is a measure of the stiffness or rigidity of a spring. It also describes how much force is needed to stretch or compress the spring by a certain amount. The spring constant is represented by the symbol k .

Q: What is the *Effective Mass* of a spring?

The effective mass of a spring is the portion of the spring's own mass that contributes to the motion and kinetic energy when the spring is oscillating.

Q: Infer the extension of your spring when a load of 300 gm is used from the load ~ extension graph.

If I add load, then show the graph increased length and also increased the time period.

Q: Does the time period of oscillation depend on the displacement from the equilibrium position? Explain

No, the time period of oscillation does not depend on the displacement from the equilibrium position.

$T = 2\pi \sqrt{m/k}$: From the formula, time period depends only on the mass and spring constant.

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Q: What happens to the time period if you keep increasing the load?

If the load attached to the spring is increased, the time of oscillation increased.

$$T = 2\pi \sqrt{\frac{m}{g}} ; T \propto \sqrt{m}$$

Q: What type of motion does the spring-block system have? Write the differential equation for such motion. Write the solution of the differential equation.

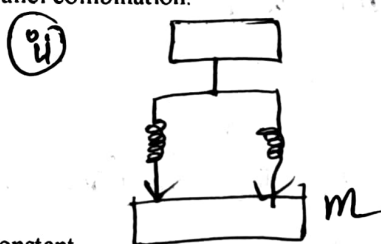
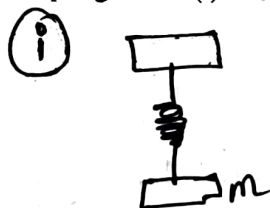
A spring-block system performs simple harmonic motion (SHM).
differential equation from Hooke's law;

$$\frac{d^2x}{dt^2} + \omega^2 x = 0 ; \omega = \sqrt{\frac{g}{m}}$$

Solution of the equation.

$$x = A \cos(\omega t + \phi)$$

Q: Suppose two spring with the same spring constant k . Draw the diagram, when these two springs are in (i) series and (ii) parallel combination.



Show that the equivalent spring constant

(i) $k_{series} = \frac{k}{2}$

$$x_1 = \frac{F}{k} , x_2 = \frac{F}{k}$$

$\therefore x = x_1 + x_2$ | if series combination

$$\Rightarrow \frac{F}{k} + \frac{F}{k}$$

$$\Rightarrow \frac{2F}{k}$$

$$\frac{F}{k_s} = \frac{2F}{k} \Rightarrow k_s = \frac{k}{2} \text{ (shown)}$$

(ii) $k_{parallel} = 2k$

$$f_1 = kx$$

$$f_2 = kx$$

if parallel,

$$F = k_p x$$

$$\therefore k_p x = 2kx \Rightarrow k_p = 2k$$

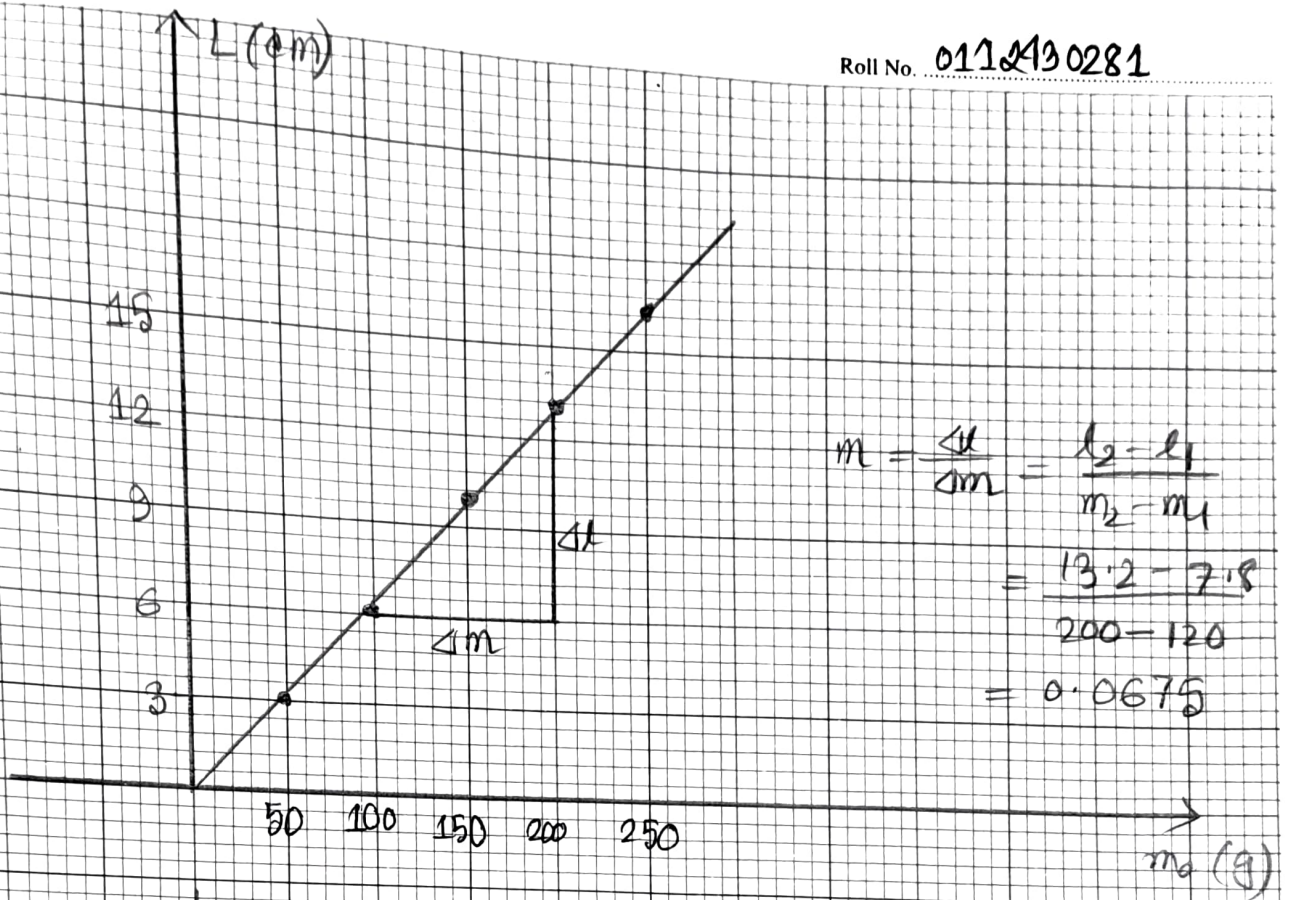
(shown)

$$F = f_1 + f_2$$

$$= kx + kx$$

$$= 2kx$$

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$$m = \frac{\Delta L}{\Delta m} = \frac{l_2 - l_1}{m_2 - m_1}$$

$$= \frac{13.2 - 7.8}{200 - 120}$$

$$= 0.0675$$

T^2 (s²)

0.25
0.5
0.75

50 100 150 200 250

m (g)